

# Commuting Cost, Labor Market Power and Misallocation: Evidence from Chinese Bridge

Yuwan Duan<sup>\*</sup>

Xiaofan Li<sup>†</sup>

Chengjun Wu<sup>‡</sup>

This version: Jan 2026

## Abstract

Bridge

**Keywords:** Markdown; China

**JEL Codes:** H77, R12, R13, D73

---

<sup>\*</sup>Central University of Finance and Economics

<sup>†</sup>Southwest University of Finance and Economics

<sup>‡</sup>Central University of Finance and Economics

# 1 Introduction

## 2 Background

## 3 Empirical Evidence

This section describes the reduced-form specifications that discipline the empirical facts and the quantitative model. The empirical exercise compares firms that receive larger commuting-time improvements from the cross-bay transport shock with firms that receive smaller improvements, before and after the opening of the bridge/tunnel system. The unit of observation is firm  $j$  in year  $t$ . The main outcomes are log employment, log wage, and log markdown:

$$y_{jt} \in \{\log L_{jt}, \log w_{jt}, \log v_{jt}\}.$$

Employment and wages are observed in the firm panel. The markdown outcome is constructed from the production-function estimation and is used as the empirical counterpart to the labor-market-power object in the model below.

The treatment variable is built from the change in commuting times between residence locations and firms. Let  $d_{zj}^0$  denote the travel time from residence location  $z$  to firm  $j$  in the no-bridge network, and let  $d_{zj}^1$  denote the corresponding travel time in the network with the bridge/tunnel links. Let  $\lambda_z$  be the normalized working-age population share of location  $z$ , with  $\sum_z \lambda_z = 1$ . Firm  $j$ 's market-access gain is

$$\Delta MA_j = \sum_z \lambda_z (d_{zj}^0 - d_{zj}^1). \quad (1)$$

A positive value means that the cross-bay links reduce the population-weighted commuting time to the firm. The main treatment indicator is

$$BigMA_j = \mathbb{1}\{\Delta MA_j \geq 0.5\}, \quad (2)$$

where the threshold is measured in minutes. The post period is defined as  $Post_t = \mathbb{1}\{t \geq 2011\}$ .

The baseline difference-in-differences specification is

$$y_{jt} = \beta_1 BigMA_j \times Post_t + \mathbf{X}_{jt}' \Gamma + \alpha_j + \lambda_t + \varepsilon_{jt}. \quad (3)$$

Firm fixed effects  $\alpha_j$  absorb time-invariant firm characteristics and the level of the treatment variable. Year fixed effects  $\lambda_t$  absorb aggregate shocks common to all firms. The control vector  $\mathbf{X}_{jt}$  includes

firm age and year-varying controls for export status, ownership type, and two-digit industry. Standard errors are clustered at the town-year level, matching the geographic aggregation of the commuting shock and allowing arbitrary correlation among firms located in the same town and year.

To examine whether the transport shock reallocates workers differently across firms with different initial wage levels, the main heterogeneous specification interacts treatment with the firm's pre-shock wage. Let

$$\tilde{w}_{j0} = \log w_{j0} - \overline{\log w_0},$$

where  $w_{j0}$  is firm  $j$ 's initial wage and the mean is recomputed within the estimation sample. The heterogeneous difference-in-differences regression is

$$\begin{aligned} y_{jt} = & \beta_1 \text{BigMA}_j \times \text{Post}_t + \beta_2 \tilde{w}_{j0} \times \text{BigMA}_j \times \text{Post}_t \\ & + \psi_t \tilde{w}_{j0} + \mathbf{X}'_{jt} \Gamma + \alpha_j + \lambda_t + \varepsilon_{jt}. \end{aligned} \quad (4)$$

The year-specific slopes  $\psi_t \tilde{w}_{j0}$  allow firms with different initial wages to follow different aggregate time paths even in the absence of differential exposure to the commuting shock. In this specification,  $\beta_1$  measures the average post-opening effect for treated firms at the mean initial wage, while  $\beta_2$  measures how this effect changes with the firm's initial wage. The employment version of equation (4) supplies two target moments used in the quantitative calibration. The wage version of the reduced-form specification supplies a third moment, the average wage response of firms with large market-access gains.

We also estimate event-study versions of equations (3) and (4). The baseline event-study equation is

$$y_{jt} = \sum_{\tau \neq \tau_0} \beta_\tau \text{BigMA}_j \times \mathbb{1}\{t = \tau\} + \mathbf{X}'_{jt} \Gamma + \alpha_j + \lambda_t + \varepsilon_{jt}, \quad (5)$$

where  $\tau_0$  is a pre-opening reference year. The heterogeneous event-study additionally includes  $\sum_{\tau \neq \tau_0} \gamma_\tau \tilde{w}_{j0} \times \text{BigMA}_j \times \mathbb{1}\{t = \tau\}$  and year-specific initial-wage slopes. These coefficients are used to inspect pre-trends and to show how the post-opening effect evolves around the shock.

Table 1: DID Regression Results

|                                   | (1)               | (2)                  | (3)                 | (4)                 | (5)                 | (6)               |
|-----------------------------------|-------------------|----------------------|---------------------|---------------------|---------------------|-------------------|
|                                   | lnemp             | lnemp                | lnmd                | lnmd                | lnw                 | lnw               |
| $BIG \times post$                 | -0.012<br>(0.025) | -0.382***<br>(0.135) | -0.056**<br>(0.023) | -0.428**<br>(0.178) | 0.065***<br>(0.022) | 0.285*<br>(0.170) |
| $BIG \times post \times \log w_0$ |                   | 0.094**<br>(0.042)   |                     | 0.114**<br>(0.054)  |                     | -0.064<br>(0.053) |
| Controls                          | Yes               | Yes                  | Yes                 | Yes                 | Yes                 | Yes               |
| Firm FE                           | Yes               | Yes                  | Yes                 | Yes                 | Yes                 | Yes               |
| Year FE                           | Yes               | Yes                  | Yes                 | Yes                 | Yes                 | Yes               |
| Observations                      | 24,350            | 16,411               | 5,066               | 5,062               | 20,025              | 16,145            |

Notes: Regressions are estimated using `reghdfe`. Standard errors are clustered at the town-sector level and reported in parentheses. \* $p < 0.10$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ .

## 4 Model Setup

This section presents the quantitative spatial model used to interpret the reduced-form evidence. Workers live in residence locations  $z \in Z$ , interpreted as streets or towns, and firms are indexed by  $j \in J$ . Each firm belongs to a sector  $s(j) \in S$ , and  $J_s$  denotes the set of firms in sector  $s$ . The mass of workers living in  $z$  is  $l_z$ , normalized so that  $\sum_z l_z = 1$ . Commuting time from  $z$  to  $j$  is  $d_{zj}$ . A firm is characterized by a wage  $w_j$ , a productivity term  $A_j$ , and a firm amenity  $a_j$ . The amenity captures non-pecuniary firm attributes that affect worker choices but do not enter the production function directly.

### 4.1 Worker Choice

Worker  $i$ , living in location  $z$ , obtains indirect utility from working at firm  $j$ ,

$$U_{izj} = \log w_j + a_j - \eta \log d_{zj} + \frac{1}{\theta} \xi_{ij}, \quad (6)$$

where  $\eta > 0$  governs the disutility of commuting time and  $\theta > 0$  governs the sensitivity of worker choices to systematic utility. The vector of idiosyncratic taste shocks follows a nested multivariate extreme-value distribution:

$$F(\xi_{i1}, \dots, \xi_{iJ}) = \exp \left\{ - \sum_{s \in S} \left[ \sum_{k \in J_s} \exp \left( - \frac{\xi_{ik}}{\sigma} \right) \right]^\sigma \right\}, \quad 0 < \sigma \leq 1. \quad (7)$$

The nest index  $s$  denotes sector. This structure allows idiosyncratic preferences over firms in the same sector to be correlated, while  $\sigma = 1$  collapses the model to the one-level logit case.

Let

$$Q_{zj} = w_j \exp(a_j) d_{zj}^{-\eta}$$

denote the systematic attractiveness of firm  $j$  to workers living in  $z$ . The probability that a worker in  $z$  chooses sector  $s$  is

$$\pi_{zs} = \frac{\left[ \sum_{k \in J_s} Q_{zk}^{\theta/\sigma} \right]^\sigma}{\sum_{r \in S} \left[ \sum_{k \in J_r} Q_{zk}^{\theta/\sigma} \right]^\sigma}. \quad (8)$$

Conditional on choosing sector  $s$ , the probability of choosing firm  $j \in J_s$  is

$$\pi_{zj|s} = \frac{Q_{zj}^{\theta/\sigma}}{\sum_{k \in J_s} Q_{zk}^{\theta/\sigma}}. \quad (9)$$

The unconditional probability of choosing firm  $j \in J_{s(j)}$  is

$$\pi_{zj} = \pi_{z,s(j)} \pi_{zj|s(j)}. \quad (10)$$

Firm  $j$ 's labor force is the sum of the workers choosing that firm across residence locations:

$$L_j = \sum_{z \in Z} l_z \pi_{zj}. \quad (11)$$

The firm-specific labor supply elasticity follows from the nested logit choice structure. For workers in residence location  $z$ , the elasticity of the choice probability with respect to firm  $j$ 's own wage is

$$\varepsilon_{zj} = \frac{\partial \log \pi_{zj}}{\partial \log w_j} = \theta \left[ \frac{1}{\sigma} + \left( 1 - \frac{1}{\sigma} \right) \pi_{zj|s(j)} - \pi_{zj} \right]. \quad (12)$$

When  $\sigma = 1$ , equation (12) becomes the one-level-logit expression  $\varepsilon_{zj} = \theta(1 - \pi_{zj})$ . Let

$$\omega_{zj} = \frac{l_z \pi_{zj}}{L_j}$$

be the share of firm  $j$ 's workforce that comes from residence location  $z$ . Firm  $j$ 's aggregate labor supply elasticity is the workforce-share-weighted average of the origin-specific elasticities:

$$\varepsilon_j = \frac{\partial \log L_j}{\partial \log w_j} = \sum_{z \in Z} \omega_{zj} \varepsilon_{zj}. \quad (13)$$

The bridge changes commuting times, and hence changes sector choices  $\pi_{zs}$ , within-sector firm choices  $\pi_{zj|s}$ , unconditional firm choices  $\pi_{zj}$ , worker-origin composition  $\omega_{zj}$ , and the labor supply elasticity  $\varepsilon_j$ .

## 4.2 Baseline Employer Groups and Wage-Ladder Poaching

The sector-nested model allows firms within the same industry to be closer substitutes, but it does not by itself make low-wage firms compete more directly with high-wage treated firms for the same workers. To capture this channel without worker-level employment histories, we augment the worker choice problem with baseline employer groups. Each firm is assigned to a group

$$g(j) = (s(j), b(j)),$$

where  $s(j)$  is the sector and  $b(j)$  is the firm's baseline wage bin. In the quantitative diagnostic below,  $b(j)$  is the decile of the baseline firm wage distribution. Let  $\bar{\ell}_g$  denote the employment-weighted mean log wage among firms in group  $g$ , and let  $s_g$  denote the sector component of group  $g$ .

Because the data do not observe worker-to-firm flows, the mass of workers initially attached to each baseline employer group is inferred from the baseline nested-logit allocation. Let

$$\gamma_{zj}^0 = \frac{l_z \pi_{zj}^0}{L_j^0} \quad (14)$$

be the share of firm  $j$ 's baseline workforce that comes from residence location  $z$ . The implied mass of workers living in  $z$  and attached to baseline employer group  $g$  is

$$M_{zg} = \sum_{j:g(j)=g} L_j^{data} \gamma_{zj}^0. \quad (15)$$

This construction preserves the observed baseline firm employment distribution while adding an approximate origin-by-employer-group state variable.

Workers from baseline group  $g$  choose among all firms after the commuting shock, but their systematic utility includes a wage-ladder term:

$$\log Q_{zgj} = \log w_j + a_j - \eta \log d_{zj} + \rho [\log w_j - \bar{\ell}_g]_+ - \chi \mathbb{1}\{s(j) \neq s_g\}. \quad (16)$$

The parameter  $\rho \geq 0$  governs how strongly workers are drawn toward firms above the wage level of their baseline employer group. The parameter  $\chi \geq 0$  allows an additional penalty for moving outside

the baseline sector group; the diagnostic exercise below sets  $\chi = 0$ . Given  $Q_{zgj}$ , group-specific sector and firm probabilities are computed with the same nested-logit formula as in equations (8)–(10), replacing  $Q_{zj}$  with  $Q_{zgj}$ . Denote the resulting probability by  $\tilde{\pi}_{zgj}$ . Firm labor in the extended model is then

$$L_j = \sum_{z \in Z} \sum_{g \in G} M_{zg} \tilde{\pi}_{zgj}. \quad (17)$$

When  $\rho = \chi = 0$ , the group-specific utilities collapse to the baseline sector-nested utility and  $\sum_g M_{zg} = l_z$ , so equation (17) nests the original labor allocation.

The wage-ladder term also changes the own-wage elasticity. Away from the kink  $\log w_j = \bar{\ell}_g$ , define

$$\lambda_{gj} = 1 + \rho \mathbb{1}\{\log w_j > \bar{\ell}_g\}.$$

The origin-group-firm elasticity is

$$\tilde{\epsilon}_{zgj} = \lambda_{gj} \theta \left[ \frac{1}{\sigma} + \left( 1 - \frac{1}{\sigma} \right) \tilde{\pi}_{zgj|s(j)} - \tilde{\pi}_{zgj} \right], \quad (18)$$

and the firm-level elasticity is

$$\epsilon_j = \sum_{z \in Z} \sum_{g \in G} \frac{M_{zg} \tilde{\pi}_{zgj}}{L_j} \tilde{\epsilon}_{zgj}. \quad (19)$$

Economically, a positive  $\rho$  makes low-wage employer groups more responsive to accessible firms higher on the wage ladder. If the bridge improves access to high-wage treated firms, low-wage control firms can lose more workers to those treated firms, generating a positive control-group wage slope in employment changes.

### 4.3 Firms and Markdown

Firm  $j$  produces with decreasing returns to labor,

$$Y_j = A_j L_j^\alpha, \quad 0 < \alpha < 1, \quad (20)$$

with product prices normalized. The firm chooses wages while recognizing that its labor supply is upward sloping. The first-order condition equates the marginal product of labor to marginal labor expenditure:

$$\alpha A_j L_j^{\alpha-1} = w_j \left( 1 + \frac{1}{\epsilon_j} \right). \quad (21)$$

Equivalently, the equilibrium wage satisfies

$$w_j = \alpha A_j L_j^{\alpha-1} \frac{\varepsilon_j}{1 + \varepsilon_j}. \quad (22)$$

The model markdown is defined as the ratio between the marginal product of labor and the wage:

$$v_j = \frac{\alpha A_j L_j^{\alpha-1}}{w_j} = 1 + \frac{1}{\varepsilon_j} > 1. \quad (23)$$

Thus, a firm with a lower labor supply elasticity has greater labor-market power and a higher markdown.

#### 4.4 Equilibrium, Inversion, and Bridge Counterfactual

Given primitives  $\{l_z, d_{zj}, A_j, a_j, s(j)\}$  and parameters  $(\alpha, \eta, \theta, \sigma)$ , an equilibrium is a vector of wages  $\{w_j\}_{j \in J}$  such that worker choices satisfy equations (8)–(10), firm labor satisfies equation (11), firm-level elasticities satisfy equation (13), and wages satisfy equation (22). Since only relative wages matter for worker choices, the numerical solution normalizes the aggregate wage bill.

In the wage-ladder extension, the equilibrium definition is identical except that firm labor and elasticities are given by equations (17) and (19). The numerical solver iterates on wages. Given a wage vector, it computes group-specific worker choices, firm labor, and firm-level elasticities, updates wages using equation (22), and repeats until the maximum log wage update is small.

For a candidate value of  $(\eta, \theta, \sigma)$ , the model maps the observed baseline economy into firm primitives in two steps. First, firm amenities are inverted from observed baseline employment and wages:

$$L_j^{data} = \sum_z l_z \pi_{zj} \left( w^{data}, d^0, a; \eta, \theta, \sigma \right), \quad \frac{1}{J} \sum_j a_j = 0. \quad (24)$$

Second, firm productivity is backed out from the wage first-order condition:

$$A_j = \kappa \frac{w_j^{data} (1 + \varepsilon_j^{data})}{\alpha (L_j^{data})^{\alpha-1} \varepsilon_j^{data}}, \quad (25)$$

where  $\kappa$  is a normalization constant. In the quantitative exercise,  $\alpha$  is fixed and  $\eta$ ,  $\theta$ , and  $\sigma$  are calibrated to match the reduced-form moments described below.

The counterfactual compares two commuting networks. The baseline network uses no-bridge commuting times  $d_{zj}^0$ . The post-opening network uses with-bridge commuting times  $d_{zj}^1$ . Holding



residence populations, firm amenities, and firm productivities fixed, we solve the model under  $d^0$  and  $d^1$ . The model-implied changes in employment, wages, and markdowns are then

$$\Delta \log L_j = \log L_j^1 - \log L_j^0, \quad \Delta \log w_j = \log w_j^1 - \log w_j^0, \quad \Delta \log v_j = \log v_j^1 - \log v_j^0.$$

For the calibration moments, the model appends the no-bridge equilibrium and the with-bridge equilibrium as a two-period firm panel and estimates the analogue of equation (4) with outcome  $\log L_j$ , firm fixed effects, period fixed effects, and period-specific initial-wage slopes. The two model coefficients on  $BigMA_j \times Post_t$  and  $\tilde{w}_{j0} \times BigMA_j \times Post_t$  are matched to their empirical counterparts. The model also estimates the corresponding two-period wage response, using outcome  $\log w_j$ , to match the reduced-form wage moment.

## 5 Calibration and Simulation

The calibration chooses the commuting-cost parameter  $\eta$ , the choice-sensitivity parameter  $\theta$ , and the sector-nest parameter  $\sigma$  so that the model reproduces the reduced-form responses of employment and wages to the bridge-induced market-access shock. The production curvature  $\alpha$  is fixed. For each candidate triple  $(\eta, \theta, \sigma)$ , the model first inverts the baseline firm amenities and firm productivities using equations (24) and (25). Holding these primitives fixed, the model then solves the no-bridge equilibrium under  $d^0$  and the post-opening equilibrium under  $d^1$ . The resulting firm-level changes in employment and wages are converted into a two-period model panel and projected onto the same treatment and wage-heterogeneity variables used in the reduced-form regressions.

The target vector contains three reduced-form coefficients:

$$\hat{m}^{RF} = \begin{pmatrix} \hat{\beta}_{L,1}^{RF} \\ \hat{\beta}_{L,2}^{RF} \\ \hat{\beta}_{w,1}^{RF} \end{pmatrix}. \quad (26)$$

The first element,  $\hat{\beta}_{L,1}^{RF}$ , is the coefficient on  $BigMA_j \times Post_t$  in the employment regression. The second element,  $\hat{\beta}_{L,2}^{RF}$ , is the coefficient on  $\tilde{w}_{j0} \times BigMA_j \times Post_t$  in the employment regression and captures whether the employment effect of the bridge differs between high- and low-wage firms. The third element,  $\hat{\beta}_{w,1}^{RF}$ , is the coefficient on  $BigMA_j \times Post_t$  in the wage regression. These three moments are computed from the reduced-form sample and then held fixed as calibration targets.

For any candidate  $(\eta, \theta, \sigma)$ , the model generates the corresponding moment vector

$$m^M(\eta, \theta, \sigma) = \begin{pmatrix} \beta_{L,1}^M(\eta, \theta, \sigma) \\ \beta_{L,2}^M(\eta, \theta, \sigma) \\ \beta_{w,1}^M(\eta, \theta, \sigma) \end{pmatrix}, \quad (27)$$

where  $\beta_{L,1}^M$  and  $\beta_{L,2}^M$  are obtained by applying the employment calibration regression to the model-generated two-period panel, and  $\beta_{w,1}^M$  is obtained by applying the wage calibration regression to model-generated wages. The estimates  $(\hat{\eta}, \hat{\theta}, \hat{\sigma})$  solve

$$(\hat{\eta}, \hat{\theta}, \hat{\sigma}) = \arg \min_{\eta, \theta, \sigma} [m^M(\eta, \theta, \sigma) - \hat{m}^{RF}]' W [m^M(\eta, \theta, \sigma) - \hat{m}^{RF}], \quad (28)$$

where the calibration uses an identity weighting matrix  $W$  and searches over a compact admissible parameter space. The two employment moments discipline the magnitude and wage heterogeneity of labor reallocation induced by the commuting-cost shock. The wage moment adds information about the equilibrium wage response, which is shaped by both labor reallocation and changes in firms' labor supply elasticities. Since commuting-time changes enter worker choice through the commuting-cost parameter, the overall choice sensitivity, and the sector-nest parameter, the wage and elasticity channels are important for separating the commuting-cost response from substitution across and within sectors.

## 5.1 Diagnostic Wage-Ladder Results

The baseline sector-nested model generates a negative control-group wage slope in the employment change, while the empirical counterpart is positive. To diagnose this sign, we compute the two-period first-difference projection

$$\Delta \log L_j = \gamma_L \tilde{w}_{j0} + \beta_{L,1} \text{BigMA}_j + \beta_{L,2} \text{BigMA}_j \tilde{w}_{j0} + u_j, \quad (29)$$

on the model-generated firm panel. The coefficient  $\gamma_L$  is the slope of employment changes with respect to baseline wages among control firms. A positive value means that low-wage control firms lose more employment, consistent with the poaching interpretation of the reduced-form pattern.

Table 2 reports the current diagnostic exercise using sector nests, ten baseline wage bins, inverted firm amenities, and

$$(\eta, \theta, \sigma, \alpha) = (0.9, 10, 0.25, 0.8), \quad \chi = 0.$$

The full-GE rows solve the wage fixed point after the commuting shock. With  $\rho = 0$ , the full-GE slope remains slightly negative. For positive  $\rho$ , the full-GE slope turns positive and rises with the strength of the wage-ladder channel.

Table 2: Wage-Ladder Poaching Diagnostic

| $\rho$ | Static $\gamma_L$ | Full-GE $\gamma_L$ | GE iterations | GE gap                |
|--------|-------------------|--------------------|---------------|-----------------------|
| 0.00   | -0.405            | -0.041             | 239           | $9.64 \times 10^{-5}$ |
| 0.25   | 1.198             | 0.325              | 112           | $9.74 \times 10^{-5}$ |
| 0.50   | 1.303             | 0.575              | 243           | $9.92 \times 10^{-5}$ |
| 1.00   | 1.191             | 0.871              | 116           | $9.75 \times 10^{-5}$ |

Notes: The table is produced by the diagnostic script `code/03_model/diagnose_poaching_groups.jl`.

Static rows hold wages fixed at baseline values after the commuting shock. Full-GE rows solve for post-shock wages using the firm first-order condition. The coefficient  $\gamma_L$  is the control-firm wage slope in equation (29).